

The diagram illustrates the sulfur cycle with the following components and reactions:

- Species:**
 - HS^- (Hydrogen sulfide)
 - $\text{S}(0)$ (Elemental sulfur)
 - SO_3^{2-} (Sulfite)
 - $\text{S}_2\text{O}_3^{2-}$ (Thiosulfate)
 - $\text{S}_4\text{O}_6^{2-}$ (Tetrathionate)
 - SO_4^{2-} (Sulfate)
 - APS** (Adenosine phosphosulfate)
- Enzymes and Pathways:**
 - Reduction of HS^- to $\text{S}(0)$:** Catalyzed by *FccAB/Sqr*.
 - Oxidation of $\text{S}(0)$ to SO_3^{2-} :** Catalyzed by *SOR* (Sulfur oxidoreductase).
 - Reduction of SO_3^{2-} to HS^- :** Catalyzed by *DsrMKJOP* (Desulfurization module).
 - Reduction of SO_3^{2-} to $\text{S}_2\text{O}_3^{2-}$:** Catalyzed by *PhsABC*.
 - Oxidation of $\text{S}_2\text{O}_3^{2-}$ to HS^- :** Catalyzed by *PhsABC*.
 - Oxidation of $\text{S}_2\text{O}_3^{2-}$ to $\text{S}_4\text{O}_6^{2-}$:** Catalyzed by *SoxYZXAB* (solid arrow) and *SoxYZXABCD* (dashed arrow).
 - Reduction of $\text{S}_4\text{O}_6^{2-}$ to SO_4^{2-} :** Catalyzed by *TetH* (solid arrow) and *SoxB* (dashed arrow).
 - Oxidation of SO_4^{2-} to $\text{S}_2\text{O}_3^{2-}$:** Catalyzed by *TsdA/DoxDA* (solid arrow) and *Ttr/Otr* (dashed arrow).
 - Reduction of SO_4^{2-} to SO_3^{2-} :** Catalyzed by *SoxYZXAB* (dashed arrow).
 - Reduction of SO_4^{2-} to APS:** Catalyzed by *Sat* (Sulfatase).
 - Oxidation of APS to SO_3^{2-} :** Catalyzed by *AprAB* (solid arrow) and *QmoABC* (dashed arrow, involving e^-).
 - Reduction of SO_3^{2-} to APS:** Catalyzed by *DsrMKJOP* (dashed arrow, involving e^-).
 - Reduction of SO_3^{2-} to $\text{S}_2\text{O}_3^{2-}$:** Catalyzed by *PhsABC* (solid arrow).
 - Reduction of SO_3^{2-} to SO_4^{2-} :** Catalyzed by *SoeABC*, *SorAB*, and *SorT* (solid arrow).
 - Reduction of SO_3^{2-} to $\text{S}(0)$:** Catalyzed by *sHdrAB1B2B3C1C2*, *DsrABCEFH*, and *Sdo* (dashed arrow).

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